

SAT Practice Module: Gemini Generated - Answers & Explanations

Date: March 7, 2026

Math Answers

1. C ($k > 5$ or $k < -5$) * **Concept:** Quadratics & The Discriminant ($b^2 - 4ac$) * **Solution:** For a quadratic equation $ax^2 + bx + c = 0$ to have **no real solutions**, the discriminant must be negative ($b^2 - 4ac < 0$). * Here, $a = k$, $b = 10$, and $c = k$. * Substitute into the formula: $10^2 - 4(k)(k) < 0$ * $100 - 4k^2 < 0$ * $100 < 4k^2$ * $25 < k^2$ * Taking the square root of both sides gives $|k| > 5$, which means $k > 5$ or $k < -5$. * **The Trap:** Many students solve $25 < k^2$ and simply get $k > 5$, forgetting that squaring a negative number (like -6) also yields a result greater than 25. Others confuse the inequality sign and choose values between -5 and 5.

2. A ($P(x) = -2(x - 8)^2 + 120$) * **Concept:** Quadratics (Vertex Form) * **Solution:** The vertex form of a parabola is $y = a(x - h)^2 + k$, where (h, k) is the vertex (maximum or minimum). * The maximum profit is 120 at $x = 8$, so the vertex is $(8, 120)$. * Equation so far: $P(x) = a(x - 8)^2 + 120$. * Use the second point $(0, -8)$ to solve for a . * $-8 = a(0 - 8)^2 + 120$ * $-8 = 64a + 120$ * $-128 = 64a$ * $a = -2$. * **The Trap:** A common mistake is assuming $a = -1$ or just picking a negative sign because the parabola opens downward, without actually calculating the specific stretch factor using the second point.

3. 369 * **Concept:** Unit Conversions (Volume/Area) * **Solution:** You must convert the volume from cubic inches to cubic centimeters before calculating mass. * Conversion factor: 1 inch = 2.54 cm. * For volume (cubic), you must **cube** the factor: $1 \text{ inch}^3 = 2.54^3 \text{ cm}^3 \approx 16.387 \text{ cm}^3$. * Total volume in cm^3 : $450 \times 16.387 = 7,374.15 \text{ cm}^3$. * Mass = Density $\times$ Volume: $0.05 \times 7,374.15 = 368.7075 \text{ kg}$. * Round to nearest whole number: 369. * **The Trap:** The most frequent error is multiplying 450 by 2.54 only once (forgetting to cube the linear conversion factor for a 3D volume), which leads to an answer of ~57.

4. A (6) * **Concept:** Function Transformations & Quadratics * **Solution:** * First, find the minimum value of $f(x) = x^2 - 14x + 55$. * The x-coordinate of the vertex is $x = -b/2a = -(-14)/2(1) = 7$. * The y-coordinate (minimum value) is $f(7) = 7^2 - 14(7) + 55 = 49 - 98 + 55 = 6$. * $g(x) = f(x + 3)$ represents a **horizontal shift** of the graph of $f(x)$ to the left by 3 units. * Horizontal shifts change the *position* of the minimum, but not the *value* of the minimum (the height). * Therefore, the minimum value of $g(x)$ is the same as $f(x)$, which is 6. * **The Trap:** Students often waste time calculating the new equation for $g(x)$ or substituting $x + 3$ incorrectly. Others might confuse the horizontal shift with a vertical shift and think the minimum changes.

Reading & Writing Answers

5. B * **Concept:** Command of Evidence (Scientific Hypothesis) * **Solution:** The researchers' hypothesis is that "biological factors" (not climatic ones) explain the bird's absence. You need an answer that presents a specific biological factor preventing the bird from living there. * Choice B identifies a "predator" (a biological factor) that exists in the empty ranges. This directly explains why the bird isn't there, supporting the

hypothesis. * **The Trap:** Choice A focuses on temperature (climatic), which contradicts the hypothesis. Choice C says the vegetation is identical; while this rules out vegetation as a problem, it doesn't positively identify the "biological factor" causing the absence as clearly as Choice B does.

6. B (are) * **Concept:** Grammar (Subject-Verb Agreement) * **Solution:** Identify the subject. The sentence says "The **works** [of contemporary artist Kehinde Wiley...] _____ distinct." * The subject is "works" (plural). * The verb must be plural: "are". * **The Trap:** The word right before the blank is "them" (referring to portraiture) or "portraiture" or "Wiley". All these are distractors. You must ignore the prepositional phrases and modifiers to find the true subject ("works").

7. C (challenge;) * **Concept:** Grammar (Boundaries & Punctuation) * **Solution:** The sentence consists of two complete independent clauses: 1. "In 1968... presented a unique engineering challenge" 2. "the rocket's internal guidance computers had to be..." * You need a strong separator (period or semicolon). * Choice C provides a semicolon. The second clause elaborates on the first, making this a correct usage. * **The Trap:** Choice A ("challenge; however") is incorrect because "however" is a conjunctive adverb. If used here to start the second clause, it would need a comma after it ("challenge; however, the..."). Furthermore, there is no logical *contrast* between the clauses; the second explains the first. Choice B creates a comma splice.

8. C * **Concept:** Rhetorical Synthesis * **Solution:** The prompt asks to explain **Saussure's specific definition of a 'sign.'** * Look for the bullet point that defines "sign": "consists of two parts: the signifier... and the signified..." * Choice C combines the names of the parts (signifier/signified) and their definitions (form/concept) into a clear statement attributed to Saussure. * **The Trap:** Choice D is close but focuses primarily on the *signifier* rather than the whole "sign." Choice A focuses on the *arbitrary relationship* rather than the definition of the components themselves.